

Supplementary Material

TARVI: Transcription-Factor Aided RNA Velocity Inference with Supervised Latent Time and Velocity-Pseudotime Blending

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This supplement gives the full component ablation that the main paper summarises. We first set out the background theory the analysis builds on—the closed-form kinetics and the velocity-graph pseudotime that TARVI uses—at a level of detail condensed in the main text, then present the ablation and the reproducibility and scope notes that follow from it. Section and table numbering carry an **S** prefix and are local to this document.

S1. Background Theory

Closed-form four-state kinetics. TARVI inherits VeloVI’s two-branch solution of the splicing ODEs, which the main text states only in words. With transcription, splicing and degradation rates α, β, γ and a per-gene switching time t_s , the induction branch starts from $u=s=0$ and evolves over $t_{\text{ind}} = t_s \rho_{i,g}$ as

$$u_{\text{ind}}(t) = \frac{\alpha}{\beta}(1 - e^{-\beta t}), \quad s_{\text{ind}}(t) = \frac{\alpha}{\gamma}(1 - e^{-\gamma t}) + \frac{\alpha}{\gamma - \beta}(e^{-\gamma t} - e^{-\beta t}), \quad (1)$$

and the repression branch starts from the induction endpoint $(u_0, s_0) = (u_{\text{ind}}(t_s), s_{\text{ind}}(t_s))$ and evolves over $t_{\text{rep}} = (t_{\text{max}} - t_s)\tau_{i,g}$ as

$$u_{\text{rep}}(t) = u_0 e^{-\beta t}, \quad s_{\text{rep}}(t) = s_0 e^{-\gamma t} - \frac{\beta u_0}{\gamma - \beta}(e^{-\gamma t} - e^{-\beta t}), \quad (2)$$

(a small ε regularises $\gamma - \beta$), with the steady-state branches as in the main paper. The RNA velocity is $v = \beta u - \gamma s$ evaluated on these curves; in TARVI α is the cell-specific, TF-regulated rate α_i , so Eqs. (1)–(2) are evaluated per cell. These curves are what the Gene-R_{spl}² metric scores, which is why the TF rate enters the kinetic-fit discussion below.

Velocity-graph pseudotime. The blending step builds its global coordinate not from expression distances but from the *velocity* transition graph. For each cell i and neighbour j in its k NN set $\mathcal{N}(i)$, a directed transition probability comes from the cosine alignment of the velocity $\mathbf{v}_i$ with the displacement $\mathbf{s}_j - \mathbf{s}_i$,

$$\pi_{ij} = \frac{\exp(\cos(\mathbf{v}_i, \mathbf{s}_j - \mathbf{s}_i)/\sigma)}{\sum_{k \in \mathcal{N}(i)} \exp(\cos(\mathbf{v}_i, \mathbf{s}_k - \mathbf{s}_i)/\sigma)}, \quad \sigma > 0, \quad (3)$$

giving a row-stochastic matrix that points downhill along the field [1]; t^{VPT} is then the diffusion pseudotime [2] from a root cell, rescaled to $[0, 1]$. Because $\Pi = [\pi_{ij}]$ is built from the *already-trained* velocities, the blend is a post-hoc step that leaves the velocity field unchanged—the property the ablation exploits.

S2. Component Ablation

We ablate each of the three components—the TF-regulated rate (TF), the velocity–time consistency loss (VTC), and the velocity-pseudotime blend (VPT)—across all five datasets (pancreas, bone marrow, forebrain, chromaffin, scEU-organoid). Table S1 reports cross-dataset means for the VeloVI baseline; VeloVI with the blend on its *own* velocity graph (VeloVI + VPT); the full model with one component removed; and full TARVI. We write $\Delta_c m$ for the change in a metric m when component c is removed from the full model, and $\delta_{\text{VPT}} m$ for the change from adding the blend to the raw VeloVI graph.

Table S1: Component ablation: cross-dataset mean performance over all five datasets. VeloVI + VPT applies the blend ($\beta=0.7$) to the unmodified VeloVI velocity graph. Best per column among configurations in bold. All metrics $\uparrow$.

Config	ICCoH	CBDiR	VelConf	PT-Spear	PT-DistCorr	PT-Cons	RootAcc	Gene- R^2_{spl}
VeloVI	0.787	0.931	0.920	0.331	0.341	0.905	0.723	0.442
VeloVI + VPT	0.787	0.931	0.920	0.714	0.724	0.986	0.658	0.442
TARVI – TF	0.693	0.911	0.902	0.759	0.783	0.984	0.653	0.461
TARVI – VTC	0.792	0.934	0.928	0.700	0.705	0.986	0.556	0.474
TARVI – VPT	0.776	0.933	0.927	0.320	0.316	0.891	0.728	0.479
TARVI (full)	0.776	0.933	0.927	0.747	0.772	0.986	0.671	0.479

The three components act on different axes. **The blend supplies the pseudotime ordering:** $\Delta_{\text{VPT}}(\text{PT-Spear}) = 0.747 - 0.320 = 0.427$, with every velocity-field metric (ICCoH, CBDiR, VelConf, Gene- R^2_{spl}) unchanged between full and –VPT—exactly as the post-hoc construction of Eq. (3) predicts. The effect is largely model-agnostic: on the raw VeloVI graph it already gives $\delta_{\text{VPT}}(\text{PT-Spear}) = 0.714 - 0.331 = 0.383$, and the gap between VeloVI + VPT and full TARVI (+0.033) is the part specific to TARVI’s velocity field. We therefore treat the blend as an orthogonal enhancement that recovers a well-ordered pseudotime from any reasonable velocity graph.

The TF rate sharpens the velocity field rather than the temporal coordinate: Δ_{TF} is positive on ICCoH (+0.083), VelConf (+0.025) and CBDiR (+0.022) but slightly negative on PT-Spear. Its kinetic-fit gain on the curves of Eqs. (1)–(2) is real but modest ($\Delta_{\text{TF}}(\text{Gene-}R^2_{\text{spl}}) = 0.018$), by design: the basal rate is warm-started from VeloVI and frozen so the TF term is an incremental correction, VeloVI’s four-state mixture already fits per-gene spliced dynamics well, and the L1 penalty keeps the weights sparse. The cell-specific rate thus behaves as a biologically grounded regulariser that improves field consistency more than it inflates R^2 . **The VTC loss improves temporal ordering and root/terminal identification:** $\Delta_{\text{VTC}}(\text{PT-Spear}) = 0.047$ and, most clearly, $\Delta_{\text{VTC}}(\text{RootAcc}) = 0.115$, with a negligible effect on the field metrics (≤ 0.016).

Finally, the blend weight β is robust. The two endpoints of the convex blend appear directly in the table— $\beta=0$ (time-head only) is TARVI – VPT (PT-Spear 0.320) and $\beta=1$ (pure velocity-graph pseudotime) is VeloVI + VPT (0.714)—and the global value $\beta=0.7$ used throughout (full TARVI, 0.747) exceeds both. The learned time and the velocity-graph pseudotime are therefore complementary, and $\beta=0.7$ is a safe interior operating point chosen once on the pooled cross-dataset mean rather than tuned per dataset.

S3. Reproducibility, Statistical Reading, and Scope

All results use a fixed random seed and a deterministic preprocessing pipeline, and we report single-run point estimates. The values in Table S1 come from a dedicated ablation run and differ

from the main-text headline numbers by ≤ 0.01 on the aggregate means; comparisons within the table are internally consistent. This ≤ 0.01 drift is the right scale for reading small differences: gaps on that order—such as TARVI versus VeloVI on the directional metrics (CBDir, VelConf)—should be read as comparable rather than as wins, whereas the pseudotime contributions above ($\Delta_{\text{VPT PT-Spear}} = 0.427$) are an order of magnitude larger.

A few limitations bound the scope of these conclusions. We did not run a multi-seed study, so we report no confidence intervals or paired significance tests; a multi-seed analysis with paired tests across datasets is the natural next step [3]. The TF prior is rebuilt per dataset by intersecting the ENCODE/ChEA databases with each gene panel; because such databases are incomplete and tissue-context dependent, genes with no reliable TF edges fall back (through the frozen basal rate and the L1 penalty) to VeloVI-like behaviour, so for lineages or organisms outside the ENCODE/ChEA scope we expect TARVI to approach the no-TF regime rather than to fail, and a systematic study of mask completeness is left to future work. Finally, VeloTrace [4], a concurrent method targeting the same velocity–trajectory gap, has no public code release that would allow a faithful re-implementation under our preprocessing and metric protocol, so we could not include it among the baseline comparisons.

References

- [1] V. Bergen *et al.*, “Generalizing RNA velocity to transient cell states through dynamical modeling,” *Nature Biotechnology*, vol. 38, pp. 1408–1414, 2020.
- [2] L. Haghverdi *et al.*, “Diffusion pseudotime robustly reconstructs lineage branching,” *Nature Methods*, vol. 13, pp. 845–848, 2016.
- [3] J. Liu *et al.*, “Comprehensive benchmarking of RNA velocity methods across single-cell datasets,” 2026.
- [4] H. Cheng *et al.*, “VeloTrace Reconciles Divergent Velocity and Trajectory in Single-cell Transcriptomics with Deep Neural ODE,” *bioRxiv*, 2026.